

GERALD OHENE AGYEKUM

+1 254 413 6830 ♦ 2251 W Tarleton St, Apt 107, Stephenville, Texas
geraldagyekum45@gmail.com ♦ www.linkedin.com/in/gerald-ohene-agyekum/

OBJECTIVE

To develop adaptive and scalable infectious disease surveillance systems that integrate Electronic Health Records (EHR), network structure, social media posts, Bayesian inference, and machine learning to improve early outbreak detection, optimize public health resource allocation, and strengthen pandemic preparedness in the United States.

EDUCATION

PhD in Computer and Statistical Sciences

University of Rhode Island.

Starting in August 2026

Master of Science in Mathematics

Tarleton State University. GPA: 4.00/4.00

Aug 2024 - May 2026

Bachelor of Technology in Health Statistics

Kumasi Technical University. 4.62/5.00 (3.87/4.00 WES Evaluated)

Sept 2019 - Aug 2023

First Class Honors

Relevant Courses

- Graduate-level: Research Analysis, Real Analysis, Statistical Models, Linear Algebra, Mathematical Modeling, Data Science, Abstract Algebra.
- Undergraduate-level: Research Methods, Stochastic Processes, Spatial Statistics, Probability, Advanced Mathematics (Calculus & Linear Algebra).

Sivas Cumhuriyet University, Turkey

Exchange Student—Erasmus Mundus Exchange Program

Oct 2021 - Feb 2022

CERTIFICATE COURSES

Data Analytics with Python

Ghana-India Kofi Annan Centre of Excellence in ICT (GI-KACE).

Dec 2023

Introduction to Machine Learning in R

Skillup by Simplilearn.

Nov 2021

RESEARCH INTERESTS

Mathematical Modeling, Spatio-Temporal Modeling, Bayesian Inference, Differential Equations, Mathematical Biology, Network Analysis, Infectious Disease Surveillance, Machine Learning, Stochastic Processes, Scientific Computing.

RESEARCH EXPERIENCE

Advisor: Dr. Jose Herrera Diestra, Tarleton State University, Stephenville, Texas, USA *Aug 2025 - Present*

“Toward a synergistic infectious disease surveillance using EHR data and network structure”.

- Developing a synergistic approach for early disease surveillance using network-based and network-free surveillance strategies.
- Incorporating a Bayesian approach to dynamically adjust the mixing parameter as the population structure changes.

Advisor: Dr. Scott Cook, Tarleton State University, Stephenville, Texas, USA *Aug 2025 - Dec 2025*

“Network-Informed Bayesian Surveillance for Epidemic Monitoring: A Simulation-Based Inference Approach”.

- Developed a Bayesian Deep Learning framework (BayesFlow) integrating network science and simulation-based inference (SBI) for epidemic surveillance.

- Achieved amortized posterior inference in < 1 second via 50,000 pre-trained simulations, vastly outperforming traditional MCMC methods.
- Used Simulation-Based Calibration (SBC) to confirm 95% credible intervals achieved 93-94% coverage across 500 independent runs.
- Engineered network-informed priors that attained 90.6% network coverage while monitoring only 1% of the population (2x efficiency gain).

Research Assistant

May 2025 - Aug 2025

Tarleton State University, Stephenville, Texas, USA

Emerging Communication Technology Use and Health Attitudes, Beliefs, and Behaviors

- Assisted in drafting the manuscript, developing survey instruments, and conducting data analysis using factorial ANOVA and moderation techniques.

Advisor: Dr. Christopher Mitchell, Tarleton State University, Stephenville, Texas, USA Feb 2025 - Present

“Mathematical Analysis of TB–COVID-19 Co-Infection: Invasion Reproduction Numbers, Competitive Dynamics, and Control Strategies”.

- Formulated a system of 11 nonlinear differential equations capturing TB–COVID-19 coinfection dynamics.

Advisor: Dr. Scott Cook, Tarleton State University, Stephenville, Texas, USA Feb 2025 - May 2025

“Predicting Diabetic Retinopathy Using Machine Learning Approaches”.

- Utilized AutoFLAML to identify the optimal algorithm for diabetic retinopathy prediction.

Advisor: Dr. Jesse Crawford, Tarleton State University, Stephenville, Texas, USA Oct 2024 - Dec 2024

“Predictive Modeling of Breast Volume Using Anthropometric Measurements”.

- Developed regression-based predictive models (stepwise and polynomial) for breast volume estimation using a comprehensive dataset collected in Ghana.

Research Assistant, Kumasi Technical University

Feb 2023 - Aug 2023

Supervisor: Prof. Michael Fosu Ofori

“Modelling the Occurrence of Maternal Mortality in Kumasi Using Zero Inflated Models”

- Performed zero-inflated regression modeling on maternal mortality data collected from 12 health facilities across 8 sub-metros in Kumasi (2018–2022).
- Identified key risk factors and determined optimal modeling strategies.

Research Assistant, Kumasi Technical University

Apr 2023 - Jul 2024

Comparative Analysis of Machine Learning Approaches for Breast Cancer Detection

Supervisor: Mr. Samuel Akwasi Adarkwa

- Compared ML algorithms for breast cancer classification using PCA and UMAP dimensionality reduction techniques.

Collaborator: Mr. Samuel Akwasi Adarkwa, Kumasi Technical University

Mar 2022 - Jan 2023

Impact of Sample Size on Multicollinearity in High-Dimensional Logistic Regression

- Conducted R simulations analyzing multicollinearity effects in high-dimensional data and drafted the manuscript for publication.

Supervisor: Dr. Esra Gulturk, Sivas Cumhuriyet University, Turkey

Dec 2021-Jan 2022

“Identifying the Risk Factors that Influence Air Quality in Sivas”.

- Analyzed Turkish air quality data to identify pollution trends and contributing factors.

PUBLICATIONS

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- Ofori, M. F., Antwi, R., **Agyekum, G. O.**, & Ofori, M. A. (2026). Assessing the Awareness and Associated Risk Factors of Cervical Cancer Among Female Students in Kumasi Technical University, Ghana. *Clinical Epidemiology and Global Health*, 102391.

- Ofori, M. F, **Agyekum, G. O**, Ofori, M. A, Adarkwa, S. A. (2025). Time-Homogeneous Markov Modeling of HIV Progression in Patients Receiving Antiretroviral Therapy Treatment in the Ashanti Region, Ghana. *Canadian Journal of Infectious Diseases and Medical Microbiology*.
- Adu, I. K., Wireko, F. A., Adarkwa, S. A., & **Agyekum, G. O**. (2024). Mathematical analysis of Ebola considering transmission at treatment centres and survivor relapse using fractal-fractional Caputo derivatives in Uganda. *Mathematical Modelling and Numerical Simulation with Applications*, 4(3), 296-334.
- **Agyekum, G. O**, Ofori, M. O, Adarkwa, S. A, Ofori, M. A (2024). A Cross-Sectional Study on the Occurrence of Maternal Mortality in Some Selected Hospitals in Kumasi Metropolis Using Zero-Inflated Negative Binomial. *Journal of Probability and Statistics*.
- **Agyekum, G. O.**, Adarkwa, S. A., & Kusi, R. Y. (2023). Impact of sample size on multicollinearity with high dimensional data in logistic regression analysis. *International Journal of Innovation and Development*, 1(3).

MANUSCRIPTS UNDER REVIEW & IN PROGRESS

- **Gerald O. Agyekum**, Christopher Mitchell. “Mathematical Analysis of TB–COVID-19 Co-Infection: Invasion Reproduction Numbers, Competitive Dynamics, and Control Strategies”. *Scientific Reports*.
- **Gerald O. Agyekum**, Michael Fosu Ofori, Michael A. Ofori, Alhassan Ismail. “A Comprehensive Analysis of Social Determinants and Clinical Predictors of Tuberculosis Treatment Success: A 5-year retrospective study from Ghana” *Health & Social Care in the Community*.
- Michael Fosu Ofori, **Gerald O. Agyekum**, Matilda Kyerewaa, Felicia O. Tettey, Michael A. Ofori. “Socio-Economic and Dietary Determinants of Maternal Malnutrition: A Cross-Sectional Study in Ghana” *Maternal & Child Nutrition*.
- **Gerald O. Agyekum**, Samuel A. Adarkwa, Frank K. Owusu, Richard Y. Kusi. “Comparative Analysis of Machine Learning Approaches for Breast Cancer Detection utilizing PCA and UMAP as Dimensionality Reduction Techniques”.

TEACHING EXPERIENCE

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| Teaching Assistant
Tarleton State University | <i>Sept 2024 - Present</i>
<i>Stephenville, Texas</i> |
| <ul style="list-style-type: none"> • College Algebra (Instructor of record) • Assist with Pre-Calculus, Plane Trigonometry, Statistics, Probability and Calculus I, II & III at the Math Achievement Center (MAC) | |
| Teaching Assistant
Kumasi Technical University | <i>Oct 2023 - Aug 2024</i>
<i>Kumasi, Ghana</i> |
| <ul style="list-style-type: none"> • Demography, Research Methods, Epidemiology, Stochastic Processes. | |
| Teaching Assistant
Sivas Cumhuriyet University | <i>Oct 2021 - Feb 2022</i>
<i>Sivas, Turkey</i> |
| <ul style="list-style-type: none"> • Assisted with course instruction, grading, and exam supervision during Erasmus exchange. | |

CONFERENCE PRESENTATIONS

- 105th Mathematical Association of America Conference, Baylor University, TX *Mar 2026*
Toward a synergistic infectious disease surveillance using EHR data and network structure.
- 104th Mathematical Association of America Conference, Prairie View A&M University, TX *Mar 2025*
Time-Homogeneous Markov Modeling of HIV Progression in Patients Receiving Antiretroviral Therapy in the Ashanti Region, Ghana.
- 7th International Conference on Applied Sciences and Technology, Kumasi Technical University *Mar 2023*
Impact of Sample Size on Multicollinearity with High-Dimensional Data in Logistic Regression Analysis.

- Attachment Seminar, Kumasi Technical University *Mar 2022*
Risk Factors for Gestational Diabetes in Pregnant Women: A Logistic Regression Analysis.

HONOURS & AWARDS

- President's List, Tarleton State University *Fall 2024, Spring 2025, Fall 2025, Spring 2026*
- Davidson Presidential Scholarship, Tarleton State University *Aug 2025*
- College of Graduate Studies Scholarship, Tarleton State University *Aug 2025*
- Encouraging Graduate Academic Progress Scholarship
Tarleton State University *Aug 2024, 2025*
- Best Graduating Student, Faculty of Applied Sciences and Technology
Kumasi Technical University *Jan 2024*
- Best Graduating Student, Department of Statistical Sciences
Kumasi Technical University *Jan 2024*
- Statistical Knowledge Contest Winner, Ghana Statistical Service *Nov 2023*
- Dean of Students Scholarship, Kumasi Technical University *Feb 2022*
- Erasmus Mundus Exchange Program, Sivas Cumhuriyet University *Oct 2021*

VOLUNTEERING

Volunteer Tutor, Student Development Project *May 2020 - Present*
Kumasi Technical University, Ghana

- Tutor students in statistical courses including regression analysis, data analysis, and experimental design.

LEADERSHIP

- **Financial Secretary** *Oct 2021 - Sept 2022*
Department of Statistical Sciences, Kumasi Technical University *Kumasi, Ghana*
- **Head of media (Graphic designer)** *Feb 2020 - Sept 2021*
Department of Statistical Sciences, Kumasi Technical University *Kumasi, Ghana*

SKILLS

Programming Languages	R, Python, Matlab, Mathematica, LaTeX
Statistical Software	SPSS, Minitab, STATA
Design Software	Adobe Photoshop.
Languages	English (fluent), Fante (fluent)

REFERENCES

Dr. Jesse Crawford
Professor
Department of Mathematics
Tarleton State University
Telephone: +1 254-968-9536
Email: JCRAWFORD@tarleton.edu

Dr. Christopher Mitchell
Assistant Professor
Department of Qualitative and Quantitative Health Sciences
University of Texas, San Antonio
Email: mitchellc4@uthscsa.edu

Dr. Michael Fosu Ofori
Associate Professor
Department of Statistical Sciences
Kumasi Technical University
Telephone: +233 24-469-5403
Email: michael.fofori@kstu.edu.gh

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